

ELITE MATHEMATICS

Pearson Edexcel International GCSE

# Modular Unit 1

## Adaptive Classified

OL-T21 · Algebraic Fractions

### With Answers

Questions with full worked solutions

■ 37 pages

# FAMILY MAP

## Algebraic Fractions

|            |                                            |                |
|------------|--------------------------------------------|----------------|
| <b>F01</b> | <b>Add or subtract algebraic fractions</b> | 6 Q   17 marks |
|------------|--------------------------------------------|----------------|

|            |                                               |                |
|------------|-----------------------------------------------|----------------|
| <b>F02</b> | <b>Multiply or divide algebraic fractions</b> | 4 Q   16 marks |
|------------|-----------------------------------------------|----------------|

|            |                                                      |                |
|------------|------------------------------------------------------|----------------|
| <b>F03</b> | <b>Substitute or transform a rational expression</b> | 6 Q   23 marks |
|------------|------------------------------------------------------|----------------|

|            |                                                              |               |
|------------|--------------------------------------------------------------|---------------|
| <b>F04</b> | <b>Simplify an algebraic fraction by factor cancellation</b> | 1 Q   3 marks |
|------------|--------------------------------------------------------------|---------------|

**Working Unit 09**

4MA1-1H Q12

ORIGINAL QUESTION

3 marks

12 Express  $\frac{5}{4} + \frac{x-3}{6x}$  as a single fraction in its simplest form.

(Total for Question 12 is 3 marks)

**Worked Solution 09**

Add algebraic fractions

**UNIT 1**  
**3 marks****Common denominator**

$$\frac{5}{4} + \frac{x-3}{6x} = \frac{15x}{12x} + \frac{2(x-3)}{12x}.$$

**Combine and simplify**

$$\frac{15x + 2x - 6}{12x} = \frac{17x - 6}{12x}, \quad x \neq 0.$$

**Final answer**

$$\frac{17x - 6}{12x}$$

**Working Unit 25**

4MA1-2H Q23

ORIGINAL QUESTION

4 marks

23  $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$  where  $p$  is an integer.

Find the value of  $p$

Show clear algebraic working.

$$p = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

**Worked Solution 25**

Simplify a division of algebraic fractions

**UNIT 1**  
**4 marks****Factorise**

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}.$$

**Multiply by the reciprocal and cancel**

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)} = \frac{4 \cdot 10}{5} = 8.$$

**Final answer**

$$p = 8$$

**Question 09**

4MA1-1H Q12

ORIGINAL QUESTION

3 marks

(b) Express  $\frac{2x+1}{4x} + \frac{7-5x}{3x}$  as a single fraction in its simplest form.

.....  
(3)

**Worked Solution 09**

Algebraic fractions

MODULAR UNIT 1

3 marks

**Use a common denominator**

$$\frac{2x+1}{4x} + \frac{7-5x}{3x} = \frac{3(2x+1) + 4(7-5x)}{12x} = \frac{31-14x}{12x}.$$

**Final answer**

$$\frac{31-14x}{12x}$$

**Question 10**

4MA1-1H Q11 demand

ORIGINAL QUESTION UNIT 1

3 marks

- 11 Express  $\frac{3}{4} + \frac{5-x}{6x}$  as a single fraction in its simplest terms.

.....

**Worked solution 10**

Single algebraic fraction

MODULAR UNIT 1

3 marks

**Demand – Single algebraic fraction**

Use a common denominator

$$\frac{3}{4} + \frac{5-x}{6x} = \frac{9x + 2(5-x)}{12x}.$$

Simplify

$$\frac{9x + 10 - 2x}{12x} = \frac{7x + 10}{12x}.$$

**Final answer** $(7x + 10)/(12x)$

**Question 31**

4MA1-2H Q24 whole

ORIGINAL QUESTION UNIT 1

5 marks

24 Solve  $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left( \frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$  .....

**Worked solution 31**

Solve a rational algebraic equation

MODULAR UNIT 1

5 marks

**whole – Solve a rational algebraic equation**

Combine the bracketed fractions

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x+3x-4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}.$$

Factor the first fraction

$$45x^3 - 80x = 5x(3x-4)(3x+4), \quad 3x^2 + x - 4 = (3x+4)(x-1).$$

Cancel common factors

The left-hand side simplifies to 20 (with the original denominator restrictions retained).

Solve the remaining equation

$$20 = \frac{4(x+2)}{5x-8} \Rightarrow 20(5x-8) = 4(x+2).$$

Find and check x

$$96x = 168 \Rightarrow x = \frac{7}{4}, \text{ which is not an excluded denominator value.}$$

**Final answer**

$$x = 7/4$$

**Original question 27**

4MA1-2H Q23 M01

MODULAR UNIT 1

4 marks

23 Show that  $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where  $n$  is an integer to be found.

Show clear algebraic working.

**Worked solution 27**

Simplify an algebraic fraction

MODULAR UNIT 1

4 marks

**M01 – Simplify an algebraic fraction****Factorise the first numerator**

$$16x^2 - 36 = 4(2x - 3)(2x + 3).$$

**Factorise the remaining terms**

$$2x^2 + 7x + 6 = (2x + 3)(x + 2), \quad x^2 - 5x - 14 = (x - 7)(x + 2).$$

**Cancel**The quotient simplifies to  $4(2x-3)$ .**Subtract the linear expression**

$$4(2x-3)-(7+8x)=-19, \text{ so } n=-19.$$

**Final answer**

$$n = -19$$

**Original question 08**

4MA1-1H Q11 Q11(b)

**ORIGINAL QUESTION****3 marks**

(b) Express  $\frac{7}{8} - \frac{x+3}{4x}$  as a single fraction in its simplest form.

.....  
(3)

**Worked solution 08**

Chapter: Algebraic Fractions

MODULAR UNIT 1

3 marks

**Q11(b) – Chapter: Algebraic Fractions**

Family: Add or subtract algebraic fractions • Variant: Linear denominators

**Method**Use denominator  $8x$ :  $\frac{7}{8} - \frac{(x+3)}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$ .*Why:  $8x$  is a common denominator for 8 and  $4x$ .***Method**Combining gives  $\frac{[7x - 2(x+3)]}{8x} = \frac{(7x - 2x - 6)}{8x}$ .*Why: Keep the subtraction outside the bracket when combining numerators.***Method**The single simplified fraction is  $\frac{(5x - 6)}{8x}$ .*Why: Collect the numerator terms.***Final answer****single fraction:**  $\frac{(5x - 6)}{8x}$

**Original question 26**

4MA1-2H Q19 Q19

**ORIGINAL QUESTION****3 marks**

19  $a = \frac{14}{3x-7}$        $x = \frac{7}{4y-3}$

Express  $a$  in the form  $\frac{py+q}{ry+s}$  where  $p, q, r$  and  $s$  are integers.

Give your answer in its simplest form.

$a = \dots\dots\dots$

**Worked solution 26**

Chapter: Algebraic Fractions

MODULAR UNIT 1

3 marks

**Q19 – Chapter: Algebraic Fractions**

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

**Method**Substitute  $x = 7/(4y - 3)$ :  $a = 14/[21/(4y - 3) - 7]$ .*Why: Replace  $x$  in the definition of  $a$ .***Method** $a = 14(4y - 3)/[21 - 7(4y - 3)] = 14(4y - 3)/(42 - 28y)$ .*Why: Combine the denominator into one fraction.***Method**Cancel the common factor 14:  $a = (4y - 3)/(3 - 2y)$ .*Why: Factor and simplify completely.***Final answer****a:**  $a = (4y - 3)/(3 - 2y)$

24 Express

$$\left( \frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

## Worked solution

January 2020 | Question 24 | 4 marks

Family: Multiply or divide algebraic fractions • Variant: Divide factorisable algebraic fractions

Method / working

1. Simplify the first bracket to  $(2x+3)/[(2x-5)(2x-3)]$ .
2. Factor the divisor numerator and denominator.
3. Multiply by the reciprocal and cancel common factors.
4. Final fraction =  $(1-3x)/[x(2x-3)^2]$ .

Final answer:  $(1-3x)/(x(2x-3)^2)$

- 12 (a) Express  $\frac{4}{x-2} - \frac{3}{x+1}$  as a single fraction.  
Give your answer in its simplest form.

(3)

## Worked solution

January 2021 | Question 12 | 3 marks

Family: Add or subtract algebraic fractions • Variant: Linear denominators

Method / working

1. Use common denominator  $(x-2)(x+1)$ .

2. Numerator =  $4(x+1)-3(x-2)=x+10$ .

3. State the simplified fraction.

Final answer:  $(x+10)/((x-2)(x+1))$

14 (a) Solve  $\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$

Show clear algebraic working.

$a = \dots\dots\dots$   
(3)

## Worked solution

January 2021 | Question 14 | 3 marks

Family: Substitute or transform a rational expression • Variant: Equation containing algebraic fractions

Method / working

1. Multiply the equation by 20 to clear denominators.
2. Expand to obtain  $21a=84$ .
3. Therefore  $a=4$ .

Final answer:  $a=4$

21 Write  $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.  
Show clear algebraic working.

# Worked solution

January 2021 | Question 21 | 4 marks

Family: Multiply or divide algebraic fractions | Variant: Multiply factorisable algebraic fractions

## METHOD / WORKING

1. Factor  $25x^2 - 64 = (5x - 8)(5x + 8)$ ,  $x^2 - 8x + 15 = (x - 5)(x - 3)$ , and  $5x^2 - 13x - 6 = (5x + 2)(x - 3)$ .
2. Factor and cancel common factors before combining.
3. Put the remaining terms over a common denominator.
4. The simplified single fraction is  $54/(5x+2)$ .

## FINAL ANSWER

 $54/(5x+2)$

21 Given that  $x = \frac{5}{9y+5}$  and that  $y = \frac{5}{5a-2}$

find an expression for  $x$  in terms of  $a$ .

Give your expression as a single fraction in its simplest form.

## Worked solution

June 2021 | Question 21 | 4 marks

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

Method / working

1. Use the printed relations  $x=5/(9y+5)$  and  $y=5/(5a-2)$ .
2. Substitute  $y$ :  $9y+5 = 45/(5a-2)+5 = [45+5(5a-2)]/(5a-2)$ .
3. Simplify the denominator to  $5(5a+7)/(5a-2)$ , then divide 5 by this fraction.
4. Therefore  $x=(5a-2)/(5a+7)$ .

Final answer:  $(5a-2)/(5a+7)$

(b) Write as a single fraction  $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$

Give your answer in its simplest form.

(2)

# Worked solution

November 2021 | Question 12 | 2 marks

Family: Add or subtract algebraic fractions | Variant: Linear denominators

## METHOD / WORKING

1. (b)

$$2. \frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$$

$$3. = \frac{20 + 24 - 27}{30x} = \frac{17}{30x}$$

**FINAL ANSWER**

$$(b) \frac{17}{30x}$$

**21** Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

## Answer reference | May-Nov 2020 | Question 21

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

|               |                                                                                                                                                                                                                                                                           |                               |   |                                                                                                                                                                      |                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 21            | $\left( \frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$                                                                                                                                                                                      |                               |   | M1 for either<br>$(3x+2)(3x-2)$ or<br>$(3x+2)(x-5)$                                                                                                                  | M2 for<br>$\frac{9x^2 - 4}{(9x^2 - 4)(x-5)} =$<br>$\frac{1}{(x-5)}$ |
|               | $\left( \frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$                                                                                                                                                                                      |                               |   | M1 for<br>$(3x+2)(3x-2)$<br><b>and</b><br>$(3x+2)(x-5)$                                                                                                              |                                                                     |
|               | E.g. of denominators<br>$(3x-2)(3x^2-13x-10)(x-1)$ or<br>$(3x-2)(3x+2)(x-5)(x-1)$ or<br>$9x^4 - 54x^3 + 41x^2 + 24x - 20$ or<br>$(3x+2)(x-5)(x-1)$ or $3x^3 - 16x^2 + 3x + 10$ or<br>$(3x-2)(x-5)(x-1)$ or $3x^3 - 20x^2 + 27x - 10$ or<br>$(x-5)(x-1)$ or $x^2 - 6x + 5$ |                               |   | M1 (indep) ft their fractions for use of a correct common denominator for 2 fractions with <b>algebraic</b> denominators<br><br>NB: fractions need not be simplified |                                                                     |
|               | $\frac{x-1-7(x-5)}{(x-5)(x-1)}$ or $\frac{x-1-7x+35}{(x-5)(x-1)}$ or<br>$\frac{x-1-7(x-5)}{x^2-6x+5}$ or $\frac{x-1-7x+35}{x^2-6x+5}$ oe                                                                                                                                  |                               |   | M1 for a <b>correct</b> fraction with a <b>correct quadratic</b> denominator – may or may not be expanded which leads to a correct answer                            |                                                                     |
|               |                                                                                                                                                                                                                                                                           | $\frac{2(17-3x)}{(x-5)(x-1)}$ | 5 | A1 accept $\frac{34-6x}{(x-5)(x-1)}$ oe; if denominator is expanded then it must be correct                                                                          |                                                                     |
| Total 5 marks |                                                                                                                                                                                                                                                                           |                               |   |                                                                                                                                                                      |                                                                     |

# Worked solution

May-Nov 2020 | Question 21 | 5 marks

Family: Substitute or transform a rational expression | Variant: Substitute a variable relation into a fraction

## METHOD / WORKING

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| Worked solution                                                                                                 |
| May-Nov 2020   Question 21   5 marks                                                                            |
| Family: Substitute or transform a rational expression   Variant: Substitute a variable relation into a fraction |
| Method                                                                                                          |
| $\frac{13x - 2x(9x^2 - 4)}{(3x^2 - 13x - 10)}$                                                                  |
| $-\frac{7}{x-1}$                                                                                                |
| Factor:                                                                                                         |
| $9x^2 - 4 = (3x - 2)(3x + 2)$                                                                                   |
| $3x^2 - 13x - 10 = (3x + 2)(x - 5)$                                                                             |
| So the first product becomes                                                                                    |
| $\frac{1}{3x-2} \times$                                                                                         |
| $\frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$                                                                              |
| Cancel common factors:                                                                                          |
| $\frac{1}{x-5}$                                                                                                 |
| Now subtract:                                                                                                   |
| $\frac{1}{x-5} - \frac{7}{x-1}$                                                                                 |
| Use the common denominator $(x-5)(x-1)$ :                                                                       |
| $\frac{(x-1-7(x-5))}{(x-5)(x-1)}$                                                                               |
| $= \frac{(x-1-7x+35)}{(x-5)(x-1)}$                                                                              |
| $= \frac{(34-6x)}{(x-5)(x-1)}$                                                                                  |
| Answer: $\boxed{\frac{(34-6x)}{(x-5)(x-1)}}$                                                                    |

## FINAL ANSWER

13 Solve  $\frac{x+3}{4} - \frac{7-x}{5} = 4.3$

Show clear algebraic working.

$x =$  .....

(Total for Question 13 is 3 marks)

## Answer reference | January 2023 | Question 13

Family: Simplify an algebraic fraction by factor cancellation • Variant: Common monomial cancellation

| Question      | Working                                                                                                                                                                                                                  | Answer | Mark | Notes                                                                                                                                                                                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13            | eg $20 \times \frac{x+3}{4} - 20 \times \frac{7-x}{5} = 20 \times 4.3$ or<br>eg $5(x+3) - 4(7-x) = 20 \times 4.3$ or<br>eg $\frac{5(x+3)}{20} - \frac{4(7-x)}{20} (= 4.3)$ or<br>eg $\frac{5(x+3) - 4(7-x)}{20} (= 4.3)$ |        | 3    | M1 For clear intention to multiply all terms by 20 (or $4 \times 5$ ) or a multiple of 20 oe<br>or<br>to express LHS as two fractions over 20 (or $4 \times 5$ ) or a multiple of 20 oe or as a single fraction with a denominator of 20 (or $4 \times 5$ ) or a multiple of 20 oe<br>if expanded numerator, allow one error |
|               | eg $5x + 15 - 28 + 4x = 4.3 \times 20$ oe<br>eg $9x - 13 = 86$<br>eg $9x = 99$                                                                                                                                           |        |      | M1 Expanding brackets and multiplying by denominator with no more than one error in total from multiplying out brackets [we must see $4.3 \times 20$ or 86 accurately]                                                                                                                                                       |
|               | <i>Working required</i>                                                                                                                                                                                                  | 11     |      | A1 dep on M1                                                                                                                                                                                                                                                                                                                 |
| Total 3 marks |                                                                                                                                                                                                                          |        |      |                                                                                                                                                                                                                                                                                                                              |

## Worked solution

January 2023 | Question 13 | 3 marks

Family: Simplify an algebraic fraction by factor cancellation • Variant: Common monomial cancellation

Official final mark-scheme pages are embedded immediately after this question.

23  $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write  $a$  in the form  $\frac{m+ny}{p(y-1)}$  where  $m$ ,  $n$  and  $p$  are integers.

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 23 is 3 marks)

# Worked solution

November 2025 | Question 23 | 3 marks

Family: Substitute or transform a rational expression | Variant: Substitute a variable relation into a fraction

## METHOD / WORKING

Worked solution

November 2025 | Question 23 | 3 marks

Family: Substitute or transform a rational expression | Variant: Substitute a variable relation into a fraction

Method

$$a = (2x+5)/(1-x), \text{quad } x = (5-2y)/(3y)$$

Substitute for x.

Numerator:

$$2x+5 = 2((5-2y)/(3y)) + 5$$

$$= (10-4y+15y)/(3y)$$

$$= (10+11y)/(3y)$$

Denominator:

$$1-x = 1 - (5-2y)/(3y)$$

$$= (3y-5+2y)/(3y)$$

$$= (5y-5)/(3y)$$

$$= (5(y-1))/(3y)$$

Therefore

$$a = ((10+11y)/(3y)) / ((5(y-1))/(3y))$$

$$= (10+11y)/(5(y-1))$$

$$\text{Answer: } d^{\frac{10+11y}{5(y-1)}}$$

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

**FINAL ANSWER**